Olympiad Test Paper — Mathematics (Class 7)

Chapter 15: Finding the Unknown

Subject	Mathematics (NCERT Class 7)
Chapter	15 — Finding the Unknown (Linear Equations)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Solve every equation by balancing / transposition and you may always check by substituting back. Most questions need two to four steps; watch for sign traps, bracket-distribution slips, and "average of speeds"-style shortcuts that don't work. Do not write on this paper — mark answers on your answer sheet.

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1. The phrase "7 less than twice a number n" is written as:

(A) 2(n − 7) (B) 2n − 7
(C) 7 − 2n (D) 2n + 7

2. "When 5 is added to thrice a number, the result is 26." As an equation this is:

(A) 3(x + 5) = 26 (B) 5x + 3 = 26
(C) 3x − 5 = 26 (D) 3x + 5 = 26

3. Solve: 4x − 9 = 11.

(A) 5 (B) 20
(C) 0.5 (D) −5

4. Solve: x/3 + 2 = 6.

(A) 24 (B) 8
(C) 12 (D) 18

5. Solve: 7x + 4 = 3x + 20.

(A) 6 (B) 2
(C) 4 (D) 3

6. Solve: 2(x − 3) = 10.

(A) 8 (B) 5
(C) 2 (D) 11

7. Solve: $3(2x + 1) = 27$.

(A) 6 (B) 9

(C) 5 (D) 4

8. Solve: $(2x - 1)/3 = 5$.

(A) 7 (B) 8

(C) 16 (D) 23

9. Which value of x satisfies $5x - 7 = 2x + 5$?

(A) 2 (B) 3

(C) 6 (D) 4

10. The sum of three consecutive integers is 72. The largest of them is:

(A) 23 (B) 25

(C) 24 (D) 21

11. A father is 4 times as old as his son, and the sum of their ages is 50 years. The son's age is:

(A) 40 (B) 12

(C) 10 (D) 8

12. Four pencils and a ₹10 sharpener together cost ₹38. The cost of one pencil (in ₹) is:

(A) 7 (B) 9

(C) 12 (D) 6

13. The length of a rectangle is 5 cm more than its breadth, and its perimeter is 50 cm. The breadth (in cm) is:

(A) 10 (B) 15

(C) 20 (D) 7.5

14. A matchstick pattern uses 4 sticks for 1 square, 7 for 2 squares, 10 for 3 squares (each extra square adds 3 sticks). How many squares can be made with exactly 31 sticks?

(A) 7 (B) 9

(C) 10 (D) 11

15. A number increased by 12 becomes three times the number. The number is:

(A) 4 (B) 6

(C) 12 (D) 3

16. Solve: $x/2 + x/3 = 10$.

(A) 6 (B) 4

(C) 24 (D) 12

17. Solve: $4(x - 1) = 2(x + 3)$.

(A) 2 (B) 7
(C) 1 (D) 5

18. Solve: $9 - 2x = 3$.

(A) 3 (B) −3
(C) 6 (D) −6

19. The sum of two consecutive even numbers is 54. The smaller number is:

(A) 24 (B) 28
(C) 26 (D) 27

20. A box holds twenty ₹1 and ₹2 coins worth ₹35 in all. The number of ₹2 coins is:

(A) 5 (B) 15
(C) 10 (D) 20

21. For which equation is $x = 3$ the solution?

(A) $2x + 1 = 5$ (B) $x - 3 = 3$
(C) $4x - 5 = 7$ (D) $x/3 = 9$

22. Solve: $x/4 - 2 = 1$.

(A) 3 (B) 4
(C) −4 (D) 12

23. Solve: $5 - 2(x - 1) = 1$.

(A) −3 (B) 3
(C) 2 (D) 4

24. Reena is 5 years older than Sam. In 3 years the sum of their ages will be 33. Sam's present age is:

(A) 11 (B) 8
(C) 16 (D) 14

25. In the sequence 3, 5, 7, 9, ... (nth term $2n + 1$), which term equals 25?

(A) the 11th (B) the 12th
(C) the 13th (D) the 24th

26. Solve: $3(x + 2) = 2(x + 5)$.

(A) 4 (B) 6
(C) 16 (D) 2

27. Solve: $(x - 3)/2 = (x + 1)/4$.

(A) 5 (B) 3
(C) 11 (D) 7

28. ₹120 is shared between A and B so that A gets twice as much as B. B's share (in ₹) is:

(A) 80 (B) 60
(C) 40 (D) 30

29. Find a number such that doubling it and adding 3 gives the same result as tripling it and subtracting 2.

(A) 5 (B) 1
(C) −5 (D) 6

30. Two angles of a triangle are equal, and the third is 30° more than each of them. Each equal angle measures:

(A) 30° (B) 80°
(C) 40° (D) 50°

31. Solve: $12 - 3x = 2x - 8$.

(A) 2 (B) −4
(C) 4 (D) 5

32. "When a number is divided by 5 and 3 is added, the result is 8." The number is:

(A) 40 (B) 25
(C) 55 (D) 15

33. The sum of three consecutive odd numbers is 81. The middle number is:

(A) 25 (B) 29
(C) 81 (D) 27

34. Solve: $2x/3 - x/2 = 1$.

(A) 3 (B) 2
(C) 6 (D) 12

35. A family bought 5 tickets for ₹155. Adult tickets cost ₹40 and child tickets ₹25. The number of child tickets bought was:

(A) 3 (B) 2
(C) 5 (D) 4

36. $x = -2$ is a solution of which equation?

(A) $3x + 1 = 5$ (B) $2x + 9 = 5$
(C) $x - 2 = 0$ (D) $4x = 8$

37. Solve: $2(3x - 1) = 4(x + 2)$.

(A) 3 (B) 10
(C) 5 (D) 1

38. A taxi charges ₹50 fixed plus ₹12 per km. A ride costing ₹158 covered how many km?

(A) 8 (B) 9
(C) 10 (D) 12

39. Five years ago, Anil was 3 times as old as Bina. Bina is now 15. Anil's present age is:

(A) 45 (B) 30

(C) 40 (D) 35

40. Solve: $x/2 + x/4 = 9$.

(A) 12 (B) 6

(C) 18 (D) 4

41. Solve: $6x - 5 = 4x + 7$.

(A) 1 (B) 6

(C) 3 (D) 11

42. The sum of two numbers is 30, and one is 6 more than the other. The larger number is:

(A) 18 (B) 12

(C) 24 (D) 15

43. An isosceles triangle has two equal sides of length x and a base of length $(x - 2)$. Its perimeter is 25. Then x equals:

(A) 7 (B) 11

(C) 9 (D) 8

44. Solve: $3x/4 = 9$.

(A) 6 (B) 3

(C) 27 (D) 12

45. "The cost of n notebooks at ₹15 each, after a ₹20 discount on the bill" is written as:

(A) $15n - 20$ (B) $15(n - 20)$

(C) $20 - 15n$ (D) $15n + 20$

46. A two-digit number has digit-sum 9, and its tens digit is twice its units digit. The number is:

(A) 36 (B) 72

(C) 81 (D) 63

47. $x = 2$ is a solution of every equation below **except**:

(A) $3x - 1 = 5$ (B) $2x + 3 = 5$

(C) $x + 4 = 6$ (D) $5x = 10$

48. Solve: $3 - (2x - 5) = 4$.

(A) 4 (B) −2

(C) 2 (D) 6

49. ₹600 is shared among A, B and C so that B gets ₹50 more than A and C gets ₹50 more than B. A's share (in ₹) is:

(A) 200 (B) 250
(C) 150 (D) 100

50. Twice a number minus 7 equals the number plus 5. The number is:

(A) 6 (B) 12
(C) 2 (D) −12

51. Solve: $(x - 5)/3 = 2$.

(A) 11 (B) 1
(C) 6 (D) 9

52. The angles of a triangle are in the ratio 1 : 2 : 3. The largest angle is:

(A) 30° (B) 60°
(C) 120° (D) 90°

53. Solve: $x/2 + 4 = x - 1$.

(A) 5 (B) 6
(C) 8 (D) 10

54. When 4 is subtracted from a number and the result is divided by 3, the answer is 5. The number is:

(A) 11 (B) 15
(C) 19 (D) 23

55. The sum of four consecutive integers is 90. The smallest of them is:

(A) 20 (B) 21
(C) 22 (D) 24

56. Solve: $2(x + 4) - 3 = x + 9$.

(A) 4 (B) 5
(C) 9 (D) 1

57. A mother is 26 years older than her daughter. In 6 years she will be 3 times as old as her daughter. The daughter's present age is:

(A) 7 (B) 5
(C) 8 (D) 10

58. A shopkeeper sold 30 fruits for ₹201, charging ₹8 per apple and ₹5 per orange. The number of apples sold was:

(A) 13 (B) 17
(C) 15 (D) 20

59. Solve: $(2x + 1)/3 = (x + 4)/2$.

(A) 5 (B) 7
(C) 8 (D) 10

60. Find the positive integer for which 3 times the integer is 8 more than twice the integer.

(A) 4 (B) 16

(C) 8 (D) 24

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 15: Finding the Unknown

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	C	31	C	41	B	51	A
2	D	12	A	22	D	32	B	42	A	52	D
3	A	13	A	23	B	33	D	43	C	53	D
4	C	14	C	24	A	34	C	44	D	54	C
5	C	15	B	25	B	35	A	45	A	55	B
6	A	16	D	26	A	36	B	46	D	56	A
7	D	17	D	27	D	37	C	47	B	57	A
8	B	18	A	28	C	38	B	48	C	58	B
9	D	19	C	29	A	39	D	49	C	59	D
10	B	20	B	30	D	40	A	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) "Twice a number" = $2n$; "7 less than" it = $2n - 7$. ($2(n - 7)$ would be "7 less than the number, then doubled".)

2. (D) "Thrice a number" = $3x$; "5 added" = $3x + 5$; "result is 26" → $3x + 5 = 26$.

3. (A) $4x - 9 = 11 \rightarrow 4x = 20 \rightarrow x = 5$.

4. (C) $x/3 + 2 = 6 \rightarrow x/3 = 4 \rightarrow x = 12$. (Dividing 6 by 3 first gives the trap 8.)

5. (C) $7x + 4 = 3x + 20 \rightarrow 7x - 3x = 20 - 4 \rightarrow 4x = 16 \rightarrow x = 4$. Check: $7 \cdot 4 + 4 = 32 = 3 \cdot 4 + 20$. ✓

6. (A) $2(x - 3) = 10 \rightarrow x - 3 = 5 \rightarrow x = 8$. Check: $2 \cdot 5 = 10$. ✓

7. (D) $3(2x + 1) = 27 \rightarrow 6x + 3 = 27 \rightarrow 6x = 24 \rightarrow x = 4$. Check: $3(8 + 1) = 27$. ✓

8. (B) $(2x - 1)/3 = 5 \rightarrow 2x - 1 = 15 \rightarrow 2x = 16 \rightarrow x = 8$. Check: $(16 - 1)/3 = 5$. ✓

9. (D) $5x - 7 = 2x + 5 \rightarrow 3x = 12 \rightarrow x = 4$. Check: $20 - 7 = 13 = 8 + 5$. ✓

10. (B) Let the integers be n, n + 1, n + 2. Sum = 3n + 3 = 72 → 3n = 69 → n = 23. Largest = n + 2 = 25. (23 is the smallest — a common trap.)

11. (C) Son = s, father = 4s. s + 4s = 50 → 5s = 50 → s = 10. (40 is the father's age.)

12. (A) 4x + 10 = 38 → 4x = 28 → x = 7.

13. (A) breadth = b, length = b + 5. Perimeter 2(b + 5 + b) = 50 → 2(2b + 5) = 50 → 2b + 5 = 25 → 2b = 20 → b = 10. (Length is 15.)

14. (C) Sticks for n squares = 3n + 1. Set 3n + 1 = 31 → 3n = 30 → n = 10. Check: 3·10 + 1 = 31. ✓

15. (B) x + 12 = 3x → 12 = 2x → x = 6.

16. (D) x/2 + x/3 = 10 → multiply by 6: 3x + 2x = 60 → 5x = 60 → x = 12. Check: 6 + 4 = 10. ✓

17. (D) 4(x − 1) = 2(x + 3) → 4x − 4 = 2x + 6 → 2x = 10 → x = 5. Check: 4·4 = 16 = 2·8. ✓

18. (A) 9 − 2x = 3 → −2x = −6 → x = 3. (Sign slips give −3.)

19. (C) n + (n + 2) = 54 → 2n + 2 = 54 → 2n = 52 → n = 26. Check: 26 + 28 = 54. ✓

20. (B) Let t = number of ₹2 coins; then 20 − t are ₹1 coins. 2t + 1·(20 − t) = 35 → t + 20 = 35 → t = 15. Check: 15·2 + 5·1 = 35. ✓

21. (C) Test x = 3: (A) 2·3 + 1 = 7 ≠ 5; (B) 3 − 3 = 0 ≠ 3; (C) 4·3 − 5 = 7 ✓; (D) 3/3 = 1 ≠ 9. So (C).

22. (D) x/4 − 2 = 1 → x/4 = 3 → x = 12.

23. (B) 5 − 2(x − 1) = 1 → 5 − 2x + 2 = 1 → 7 − 2x = 1 → 2x = 6 → x = 3. (The trap is writing 5 − 2x − 2; the bracket flips the sign of −1 to +2.) Check: 5 − 2(2) = 1. ✓

24. (A) Now: Sam = s, Reena = s + 5. In 3 years: (s + 3) + (s + 5 + 3) = 33 → 2s + 11 = 33 → 2s = 22 → s = 11. Check: in 3 years 14 + 19 = 33. ✓

25. (B) 2n + 1 = 25 → 2n = 24 → n = 12. So the 12th term.

26. (A) 3(x + 2) = 2(x + 5) → 3x + 6 = 2x + 10 → x = 4. Check: 3·6 = 18 = 2·9. ✓

27. (D) (x − 3)/2 = (x + 1)/4 → cross-multiply: 4(x − 3) = 2(x + 1) → 4x − 12 = 2x + 2 → 2x = 14 → x = 7. Check: 4/2 = 2 = 8/4. ✓

28. (C) B = b, A = 2b. 2b + b = 120 → 3b = 120 → b = 40. (A gets 80.)

29. (A) 2x + 3 = 3x − 2 → 3 + 2 = 3x − 2x → x = 5. Check: 2·5 + 3 = 13 = 3·5 − 2. ✓

30. (D) $x + x + (x + 30) = 180 \rightarrow 3x + 30 = 180 \rightarrow 3x = 150 \rightarrow x = 50$. Each equal angle = $50°$ (third = $80°$). ✓

31. (C) $12 - 3x = 2x - 8 \rightarrow 12 + 8 = 2x + 3x \rightarrow 20 = 5x \rightarrow x = 4$. Check: $12 - 12 = 0 = 8 - 8$. ✓

32. (B) $x/5 + 3 = 8 \rightarrow x/5 = 5 \rightarrow x = 25$.

33. (D) Three consecutive odd numbers average to the middle one, so middle = $81 \div 3 = 27$. (Numbers: 25, 27, 29.) ✓

34. (C) $2x/3 - x/2 = 1 \rightarrow$ multiply by 6: $4x - 3x = 6 \rightarrow x = 6$. Check: $12/3 - 6/2 = 4 - 3 = 1$. ✓

35. (A) Child tickets = c, adult = 5 − c. $40(5 - c) + 25c = 155 \rightarrow 200 - 40c + 25c = 155 \rightarrow 200 - 15c = 155 \rightarrow 15c = 45 \rightarrow c = 3$. Check: 3 child (₹75) + 2 adult (₹80) = ₹155. ✓

36. (B) Test $x = -2$: (B) $2(-2) + 9 = 5$ ✓. The others: (A) $-5 \neq 5$, (C) $-4 \neq 0$, (D) $-8 \neq 8$. So (B).

37. (C) $2(3x - 1) = 4(x + 2) \rightarrow 6x - 2 = 4x + 8 \rightarrow 2x = 10 \rightarrow x = 5$. Check: $2 \cdot 14 = 28 = 4 \cdot 7$. ✓

38. (B) $50 + 12k = 158 \rightarrow 12k = 108 \rightarrow k = 9$.

39. (D) Bina now = 15, so 5 years ago Bina = 10 and Anil = $3 \cdot 10 = 30$. Anil now = $30 + 5 = 35$.

40. (A) $x/2 + x/4 = 9 \rightarrow$ multiply by 4: $2x + x = 36 \rightarrow 3x = 36 \rightarrow x = 12$. Check: $6 + 3 = 9$. ✓

41. (B) $6x - 5 = 4x + 7 \rightarrow 2x = 12 \rightarrow x = 6$. Check: $36 - 5 = 31 = 24 + 7$. ✓

42. (A) Numbers x and $x + 6$. $x + (x + 6) = 30 \rightarrow 2x = 24 \rightarrow x = 12$. Larger = $x + 6 = 18$.

43. (C) $x + x + (x - 2) = 25 \rightarrow 3x - 2 = 25 \rightarrow 3x = 27 \rightarrow x = 9$. Check: $9 + 9 + 7 = 25$. ✓

44. (D) $3x/4 = 9 \rightarrow 3x = 36 \rightarrow x = 12$.

45. (A) Cost of n notebooks = $15n$; "after a ₹20 discount" = $15n - 20$.

46. (D) units = u, tens = 2u. Digit-sum $2u + u = 9 \rightarrow 3u = 9 \rightarrow u = 3$, tens = $6 \rightarrow$ the number is 63. Check: $6 + 3 = 9$ and $6 = 2 \cdot 3$. ✓

47. (B) Test $x = 2$: (A) $5 = 5$ ✓, (B) $2 \cdot 2 + 3 = 7 \neq 5$ ✗, (C) $6 = 6$ ✓, (D) $10 = 10$ ✓. The exception is (B).

48. (C) $3 - (2x - 5) = 4 \rightarrow 3 - 2x + 5 = 4 \rightarrow 8 - 2x = 4 \rightarrow 2x = 4 \rightarrow x = 2$. Check: $3 - (4 - 5) = 3 - (-1) = 4$. ✓

49. (C) $A = a$, $B = a + 50$, $C = a + 100$. Sum $= 3a + 150 = 600 \rightarrow 3a = 450 \rightarrow a = 150$. Check: $150 + 200 + 250 = 600$. ✓

50. (B) $2x - 7 = x + 5 \rightarrow x = 12$. Check: $2 \cdot 12 - 7 = 17 = 12 + 5$. ✓

51. (A) $(x - 5)/3 = 2 \rightarrow x - 5 = 6 \rightarrow x = 11$.

52. (D) Angles x, $2x$, $3x$. $x + 2x + 3x = 180 \rightarrow 6x = 180 \rightarrow x = 30$. Largest $= 3x = 90°$.

53. (D) $x/2 + 4 = x - 1 \rightarrow 4 + 1 = x - x/2 \rightarrow 5 = x/2 \rightarrow x = 10$. Check: $10/2 + 4 = 9 = 10 - 1$. ✓

54. (C) $(x - 4)/3 = 5 \rightarrow x - 4 = 15 \rightarrow x = 19$.

55. (B) $n + (n+1) + (n+2) + (n+3) = 90 \rightarrow 4n + 6 = 90 \rightarrow 4n = 84 \rightarrow n = 21$. Check: $21 + 22 + 23 + 24 = 90$. ✓

56. (A) $2(x + 4) - 3 = x + 9 \rightarrow 2x + 8 - 3 = x + 9 \rightarrow 2x + 5 = x + 9 \rightarrow x = 4$. Check: $2 \cdot 8 - 3 = 13 = 4 + 9$. ✓

57. (A) daughter = d, mother = $d + 26$. In 6 years: $(d + 26 + 6) = 3(d + 6) \rightarrow d + 32 = 3d + 18 \rightarrow 14 = 2d \rightarrow d = 7$. Check: mother 33; in 6 years $39 = 3 \cdot 13$. ✓

58. (B) apples = a, oranges = $30 - a$. $8a + 5(30 - a) = 201 \rightarrow 8a + 150 - 5a = 201 \rightarrow 3a = 51 \rightarrow a = 17$. Check: $17 \cdot 8 + 13 \cdot 5 = 136 + 65 = 201$. ✓

59. (D) $(2x + 1)/3 = (x + 4)/2 \rightarrow$ cross-multiply: $2(2x + 1) = 3(x + 4) \rightarrow 4x + 2 = 3x + 12 \rightarrow x = 10$. Check: $21/3 = 7 = 14/2$. ✓

60. (C) $3x = 2x + 8 \rightarrow x = 8$. Check: $3 \cdot 8 = 24 = 2 \cdot 8 + 8$. ✓

Solutions complete: 60 / 60.